

# Quick Notes

## Extension of ABC Theorem (9.26.2026):

The ABC theorem applies specifically to a self-adjoint Hamiltonian on Euclidean configuration space (an operator  $H = -\Delta + V(x)$  on  $L^2(\mathbb{R}^n)$ ), with  $V$  satisfying an appropriate dilation-analyticity condition. The following is a paper wherein Kelvin extends this theorem to a more general setting. It's terse, and I don't think it has a use for us, but I put it here in case.

<https://arxiv.org/abs/1003.2538>

## Notes on ABC Theorem (9.26.2026):

### Inner Products

I will likely move these to phd-reference. Inner product space is a vector space  $V$  over the field  $F$  together with an inner product  $\langle \cdot, \cdot \rangle : V \times V \rightarrow F$ , satisfying the following properties:

1. Conjugate symmetry:  $\langle x, y \rangle = \overline{\langle y, x \rangle}$ .
2. Linearity in the first argument:  $\langle ax + by, z \rangle = a\langle x, z \rangle + b\langle y, z \rangle$ .
3. Positive-definiteness: if  $x$  is not zero, then  $\langle x, x \rangle > 0$ .

Properties 1 and 2 imply that an inner product must be conjugate-linear in the second argument, making it a sesquilinear form. When the field is  $F = \mathbb{R}$ , conjugate symmetry reduces to symmetry, and sesquilinearity reduces to bilinearity. Thus, over  $F = \mathbb{C}$ , an inner product is a positive-definite, conjugate-symmetric, sesquilinear form, and over  $F = \mathbb{R}$ , it reduces to a positive-definite, symmetric, bilinear form. The c-product is not technically an inner product because  $(\psi, \phi)_c = (\phi, \psi)_c$ ; i.e., it's symmetric, not conjugate symmetric. Thus, it's a symmetric, bilinear form over the space. The adjoint of an operator  $H$ , often denoted  $H^\dagger$ , under a specific inner product  $\langle \cdot, \cdot \rangle$  is defined by  $\langle H^\dagger \psi, \phi \rangle = \langle \psi, H \phi \rangle \forall \psi, \phi \in V$ .

### Left / Right Eigenvectors

For an operator  $H : V \rightarrow V$ , a right eigenvector  $|\psi_R\rangle$  is defined by  $H|\psi_R\rangle = E|\psi_R\rangle$ . The corresponding left eigenvector is defined by conjugate transposing the equation: